

DESCRIPTIVE STATISTICS

- **Mean** = sum/n (pulled by outliers). **Median** = middle value (robust). **Mode** = most frequent.
- **Variance** (sample) = $\sum(x - \text{mean})^2 / (n - 1)$; population divides by n. **SD** = $\sqrt{\text{variance}}$, same units as x.
- **IQR** = $Q3 - Q1$. Outlier rule: below $Q1 - 1.5 \cdot \text{IQR}$ or above $Q3 + 1.5 \cdot \text{IQR}$.
- **z-score** = $(x - \text{mean}) / \text{SD}$. Skew: right-skewed (income) means mean > median.
- **Covariance** = $\sum(x - \text{mx})(y - \text{my}) / (n - 1)$. **Correlation r** = $\text{cov}(\text{sx}, \text{sy})$, from -1 to 1: linear association only, units-free. $r = 0$ does not mean independent.
- Dividing by $n - 1$ (Bessel) corrects for estimating the mean from the same data.
- Robust spread: **MAD** = median $|x - \text{median}|$.

TRAP Correlation is not causation; confounders and reverse causation. Simpson's paradox: a trend reverses when groups are combined.

PROBABILITY RULES

- $0 \leq P(A) \leq 1$; $P(\text{not } A) = 1 - P(A)$. "At least one" = $1 - P(\text{none})$.
- Addition: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.
- Multiplication: $P(A \text{ and } B) = P(A) P(B | A)$. Independent: $P(A \text{ and } B) = P(A) P(B)$.
- Conditional: $P(A | B) = P(A \text{ and } B) / P(B)$.
- Total probability: $P(B) = \sum P(B | A_i) P(A_i)$.
- **Mutually exclusive** (cannot both happen) is not **independent**; exclusive events with $P > 0$ are dependent.
- Counting: $n!$ orderings; $nPr = n! / (n - r)!$; $nCr = n! / (r!(n - r)!)$.
- Expectation: $E[X + Y] = E[X] + E[Y]$ always (linearity). $\text{Var}(aX + b) = a^2 \text{Var}(X)$; $\text{Var}(X + Y) = \text{Var } X + \text{Var } Y$ if independent.

HANDY FACTS AND INEQUALITIES

- Markov: $P(X \geq a) \leq E[X] / a$ ($X \geq 0$). Chebyshev: $P(|X - \mu| \geq k\sigma) \leq 1/k^2$, any distribution (75% within 2σ).
- Percentiles: median = 50th, quartiles 25/75th; for the normal, the 97.5th percentile is $\mu + 1.96\sigma$.
- Skewness: sign gives the long tail; kurtosis: heavy tails. Log-transform right-skewed positives.
- Regression slope = $r(\text{sy}/\text{sx})$. Correlation is symmetric; regression is not.
- Law of total expectation: $E[X] = \sum E[X | A] P(A)$. Variance of a sum of n iid = $n\sigma^2$.
- Independence of events: $P(A | B) = P(A)$. Conditional independence differs from marginal independence.

- Expected value of a game = $\sum \text{value} \times \text{probability}$; decide with EV, mention variance.

BAYES' THEOREM

$P(H | E) = P(E | H) P(H) / P(E)$, posterior = likelihood $\times$ prior / evidence.

- Worked example: prevalence 1%, sensitivity 99%, false-positive 5%. $P(+)$ = $0.99 \cdot 0.01 + 0.05 \cdot 0.99 = 0.0594$. $P(\text{disease} | +)$ = $0.0099 / 0.0594 = 16.7\%$.
- Natural frequencies: of 10,000 people, 100 sick (99 test positive), 9,900 healthy (495 test positive): 99 of 594.
- Base-rate neglect: ignoring a small prior. Sensitivity = $TP / (TP + FN)$; specificity = $TN / (TN + FP)$.
- Odds form: posterior odds = prior odds $\times$ likelihood ratio.

CLASSIC PUZZLES

Puzzle	Answer
Two dice sum to 7	$6/36 = 1/6$
Two children, at least one girl, both girls	$1/3$
Birthday: 23 people share a birthday	> 50% (253 pairs)
Monty Hall	switch wins 2/3
Flips to first head	$E = 2$
Flips to HH / HT	$6 / 4$
Rolls to first six	6
Coupon collector, n types	$n(1 + 1/2 + \dots + 1/n)$

Method: define the sample space, count, then check by simulation.

DISTRIBUTIONS

Distribution	Mean	Variance	Use
Bernoulli(p)	p	$p(1 - p)$	one yes/no trial
Binomial(n, p)	np	$np(1 - p)$	successes in n trials
Geometric(p)	$1/p$	$(1 - p)/p^2$	trials to first success
Poisson(λ)	λ	λ	events per interval, rare and independent
Uniform(a, b)	$(a + b)/2$	$(b - a)^2/12$	equally likely
Normal(μ, σ^2)	μ	σ^2	sums and averages, errors
Exponential(λ)	$1/\lambda$	$1/\lambda^2$	waiting time, memoryless
Beta(α, β)	$\alpha/(\alpha + \beta)$		probability of a probability

- Binomial with large n, small p $\rightarrow$ Poisson(np). Normal approx when np and $n(1 - p) > 10$.
- Normal: 68% within 1σ , 95% within 2σ (1.96σ), 99.7% within 3σ . z crit: 1.645 (90%), 1.96 (95%), 2.576 (99%).
- t has heavier tails than normal, approaches it as df grows. Chi-square = sum of squared normals. F = ratio of variances.
- Log-normal / power law for incomes, latencies: report medians and percentiles.

PROBABILITY RECIPES

- Binomial: $P(X = k) = C(n, k) p^k (1 - p)^{n-k}$. Poisson: $P(X = k) = e^{-\lambda} \lambda^k / k!$.
- Draw without replacement (cards): multiply conditional probabilities, or use combinations: $P(2 \text{ aces}) = C(4, 2) / C(52, 2) = 1/221$.
- At least one success in n tries: $1 - (1 - p)^n$ (four rolls, a six: $1 - (5/6)^4 = 51.8\%$).
- Expected value of a die = 3.5; of a sum = sum of expectations. Gambler's game: $EV = \sum \text{payout} \times \text{probability}$.
- Conditional on a total: "given sum = 7, $P(\text{first die} = 3) = 1/6$ by symmetry.
- Random variable transformations: $E[aX + b] = a E[X] + b$. Sum of independent normals is normal.
- Simulate when stuck: `np.random.binomial`, then count frequencies.

CLT AND LLN

- **LLN**: the sample mean converges to the true mean as n grows.
- **CLT**: the mean of n independent samples is approximately normal with mean μ and **standard error SE** = $\sigma / \sqrt{n}$, whatever the data's shape (finite variance), usually $n \geq 30$, more for heavy skew.
- $4 \times$ the data halves the error. It is about **means**, not about the data becoming normal.
- SE of a proportion = $\sqrt{p(1 - p)/n}$. SE of a difference = $\sqrt{SE1^2 + SE2^2}$.
- Law of rare events, bootstrap and the normal approximation all lean on the CLT.

CONFIDENCE INTERVALS

- Mean: sample mean $\pm t \cdot s / \sqrt{n}$ (z when n is large or σ known). Proportion: $p \pm 1.96 \cdot \sqrt{p(1-p)/n}$.
- 95% CI meaning:** if we repeated the sampling and the method many times, 95% of the intervals would cover the true value. Not "95% probability this interval holds it".
- Wider with more variance, higher confidence, smaller n. Margin of error = critical value $\times$ SE.
- Report the interval, not just the p-value: it shows effect size and uncertainty.

HYPOTHESIS TESTING

- State H_0 (no effect) and H_1 .
 - Choose alpha (0.05) and the test.
 - Compute the statistic and **p-value**.
 - Decide and interpret.
- p-value** = P(data this extreme or more | H_0 true). Not P(H_0 true), not the effect size.
 - Type I** (false positive, rate α): reject a true H_0 . **Type II** (miss, rate β): fail to reject a false H_0 . **Power** = $1 - \beta$, usually 80%.
 - Power rises with larger n, larger effect, lower variance, higher α . One-sided tests need a prior direction.
 - Statistical vs practical significance: huge n makes tiny effects significant. Report effect size (difference, Cohen's d, odds ratio).
 - Not significant $\neq$ no effect: low power gives wide intervals.

TRAP Peeking at results and stopping at the first $p < 0.05$ inflates false positives.

WHICH TEST?

Data and question	Test
Mean vs known value	one-sample t
Two independent means	Welch t-test (unequal variances)
Same units before/after	paired t-test
3+ group means	ANOVA, then post-hoc (Tukey)
Two proportions	two-proportion z-test / chi-square
Two categoricals	chi-square of independence; Fisher if counts < 5
Fit to a distribution	chi-square goodness of fit, KS
Two samples, non-normal / ordinal	Mann-Whitney U; paired: Wilcoxon
3+ groups, non-normal	Kruskal-Wallis
Linear association	Pearson r; rank-based: Spearman
Two distributions differ	two-sample KS

- t statistic = (sample mean - μ_0) / (s / $\sqrt{n}$). Two proportions: $z = (p_1 - p_2) / \sqrt{p(1-p)(1/n_1 + 1/n_2)}$ with pooled p.
- Assumptions: independence, roughly normal means (CLT), equal variances for classic t/ANOVA. Heavy tails: bootstrap or transform.

SAMPLE SIZE (PER GROUP)

- Means: $n \sim 16\sigma^2/\delta^2$ (80% power, $\alpha = 0.05$). Proportions: $n \sim 16p(1-p)/\delta^2$.
- Example: baseline 10%, detect +1pp: $16 \cdot 0.09/0.0001 \sim 14,400$ per group.
- Halving the detectable effect quadruples n. Duration = total n / daily eligible traffic, in whole weeks.

MULTIPLE TESTING

- m tests at $\alpha = 0.05$: P(at least one false positive) = $1 - 0.95^m$ (20 tests: 64%).
- Bonferroni:** use α/m (controls family-wise error; strict). **Benjamini-Hochberg:** controls the false discovery rate; rank p-values, keep $p(i) \leq (i/m) \cdot q$.
- Pre-register one primary metric; treat post-hoc segment hits as hypotheses.

```
from scipy import stats
stats.ttest_ind(a, b, equal_var=False)
stats.ttest_rel(a, b)
stats.chi2_contingency(table)
stats.mannwhitneyu(a, b);
stats.ks_2samp(a, b)
stats.pearsonr(x, y);
stats.norm.ppf(0.975)
import statsmodels.stats.multitest as mt
mt.multipletests(p, method="fdr_bh")
from statsmodels.stats import power
power.TTestIndPower().solve_power(
    effect_size=0.2, power=0.8,
    alpha=0.05)
```

EFFECT SIZES AND ASSUMPTIONS

Quantity	Formula / use
Absolute lift	$p_{\text{treat}} - p_{\text{control}}$
Relative lift	$(p_{\text{treat}} - p_{\text{control}}) / p_{\text{control}}$
Cohen's d	difference in means / pooled SD (0.2 small, 0.5 medium, 0.8 large)
Odds ratio	$[p/(1-p)]_{\text{treat}} \div \text{control}$
R^2	$1 - SS_{\text{res}} / SS_{\text{total}}$

- Check normality with a histogram or Q-Q plot, equal variances with Levene's test, independence from the design (clustered users break it).
- Violations: transform (log), use Welch, use non-parametric tests, or bootstrap.
- Paired data: analyse differences; clustered data: aggregate to the randomisation unit.

OTHER ESSENTIALS

- Sampling:** simple random, stratified (represent subgroups), cluster. Biases: selection, survivorship, non-response.
- Bootstrap:** resample with replacement, recompute the statistic, take percentiles for a CI. **Permutation test:** shuffle labels.
- MLE:** choose parameters maximising the likelihood; for a normal it gives the sample mean and variance (divides by n). **MAP** adds a prior.
- Bayesian vs frequentist:** posterior = prior $\times$ likelihood gives probability statements about parameters; credible interval is a probability statement, a confidence interval is a procedure property.
- Linear algebra:** eigenvector keeps direction under a matrix; PCA = eigenvectors of the covariance matrix; SVD = $U \Sigma V^T$.
- Calculus:** the gradient points uphill; gradient descent steps against it; chain rule powers backprop.
- Regression: R^2 = explained variance share; coefficient t-test = estimate/SE.

SAY IT IN 20 SECONDS

- p-value:** "how surprising this data is if nothing is going on".
- CI:** "a range from a procedure that covers the truth 95% of the time".
- CLT:** "averages are roughly normal with SD $\sigma/\sqrt{n}$ ".
- Power:** "the chance to detect a real effect of this size".
- Bayes:** "update the prior with the likelihood; never forget the base rate".
- Definition, tiny example, one caution.